📊 Marketing Spend Waste Detector

End-to-End Data Analytics Project (Python | SQL | Power BI)

🧠 1. Project Overview

In a high-growth ecommerce environment, marketing budgets are often distributed across multiple channels without clear visibility into performance efficiency. This project aims to identify **inefficient marketing spend, detect loss-making campaigns, and optimize budget allocation** using data-driven insights.

The analysis integrates **Python for data preparation, SQL for business querying, and Power BI for visualization**, simulating a real-world analytics workflow.

🎯 2. Business Objectives

- Identify **channels and campaigns with negative ROI**
- Detect **high-spend, low-return marketing efforts**
- Analyze **customer drop-off behavior**
- Provide actionable recommendations for **budget optimization**
- Enable decision-making through an **interactive dashboard**

📂 3. Dataset Description

The dataset consists of ~2000 rows of simulated ecommerce marketing data with the following key attributes:

- Marketing Channels (Facebook Ads, YouTube Ads, etc.)
- Campaign-level performance
- Financial metrics (Spend, Revenue, Profit)
- Funnel metrics (Impressions, Clicks, Conversions)
- Customer behavior indicators (CTR, CPC, Ratings)

The dataset intentionally includes:

- Missing values
- Outliers
- Inconsistent profit calculations

- Loss-making campaigns

This ensures realistic, business-like complexity.

🐍 4. Data Processing (Python)

Key Steps:

✓ Data Cleaning

- Handled missing values using median imputation
- Fixed inconsistent data types
- Removed critical null rows

✓ Feature Engineering

- Recalculated:
  - Profit = Revenue – Cost
  - ROI = Profit / Marketing Spend

✓ Created Analytical Flags:

- High_Spend_Low_Conversion
- Negative_ROI
- Zero_Clicks

These flags enabled identification of **hidden inefficiencies**.

✓ Exploratory Analysis:

- ROI distribution showed skewness (few high performers, many average)
- Scatter analysis revealed:

  High spend does not guarantee high revenue

🗄️ 5. Business Analysis (SQL)

SQL was used to extract **decision-level insights**.

🔍 Channel Performance Analysis

- **Worst Channel:** YouTube Ads (lowest ROI)
- **Best Channel:** Facebook Ads (highest ROI)

👉 Insight:

Significant budget is being allocated to underperforming channels.

🔍 Campaign-Level Waste Detection

- Identified campaigns with:
  - High spend
  - Negative profit
  - Low ROI

👉 Insight:

Several campaigns were actively burning money but remained unnoticed due to aggregated reporting.

🔍 Funnel Analysis (Simplified)

- Email → Lowest CTR (poor engagement)
- YouTube Ads → Lowest Conversion Rate

👉 Insight:

Different stages of funnel failure exist across channels.

📊 6. Dashboard Development (Power BI)

An interactive dashboard was built to visualize insights.

🔹 KPI Section (Executive View)

- Total Spend

- Total Revenue
- Total Profit
- Average ROI
- % Negative ROI

🔹 Channel Performance

- ROI comparison across channels
- Spend vs Revenue imbalance detection

🔹 Waste Detection Table

- Campaign-level breakdown
- Conditional formatting:
  - Red → Loss
  - Green → Profit

🔹 Scatter Plot Analysis

- Visual identification of:
  - High spend, low revenue (waste zone)
  - Low spend, high revenue (opportunity zone)

🔹 Interactive Filters (Slicers)

- Region
- Channel
- Campaign
- Date

🧠 7. Key Insights

- A significant portion of campaigns generate **negative ROI**, indicating marketing inefficiency
- YouTube Ads consistently underperform despite investment
- Email campaigns suffer from **low engagement (CTR)**

- Some campaigns consume high budgets but generate **minimal conversions**
- Revenue is not linearly proportional to marketing spend

💡 8. Business Recommendations

- Reallocate budget from low ROI channels to high-performing ones (e.g., Facebook Ads)
- Optimize or discontinue consistently loss-making campaigns
- Improve email marketing strategies (targeting + content)
- Enhance landing pages to improve conversion rates
- Focus on data-driven budget allocation instead of uniform distribution

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🚀 9. Tools & Technologies

- Python (Pandas, Matplotlib) → Data Cleaning & Analysis
- SQL → Business Querying
- Power BI → Visualization & Dashboarding

🏁 10. Conclusion

This project demonstrates how raw marketing data can be transformed into actionable business insights through a structured analytics workflow.

By combining **technical skills with business understanding**, the analysis highlights inefficiencies, supports strategic decisions, and showcases real-world data analyst capabilities.

📌 Author

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